

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I P.Kalyani(192472199) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:P.Kalyani

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

QUESTION 1

Hospital Doctor Scheduling using Backtracking Search

Problem Understanding

Problem Statement

The hospital must assign three doctors (D1, D2, D3) to three different shifts (Morning, Afternoon, Night) while ensuring that all scheduling constraints are satisfied. Since every assignment depends on several restrictions, this problem is modeled as a Constraint Satisfaction Problem (CSP).

Identified Constraints

Constraint ID	Constraint
C1	D1 cannot work Night shift
C2	D2 must work before D3 (Morning < Afternoon < Night)
C3	Only one doctor can be assigned to one shift
C4	D3 cannot work Morning shift
C5	Every doctor must be assigned exactly one shift

Observation: Every constraint must be satisfied simultaneously to obtain a valid schedule.

Constraint Analysis

Constraint Priority

Priority	Constraint	Reason
High	One doctor per shift	Prevents duplicate assignments
High	Every doctor gets one shift	Completes scheduling
Medium	D2 before D3	Determines assignment order
Medium	D1 not Night	Restricts D1's choices
Medium	D3 not Morning	Restricts D3's choices

Analysis

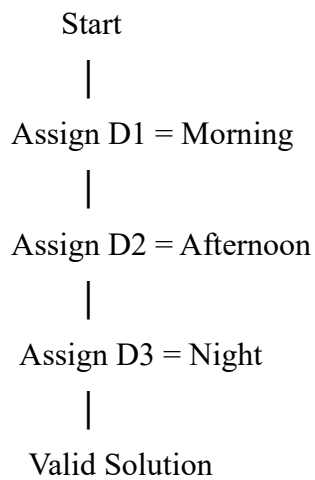
The scheduling process begins by assigning doctors one at a time. After every assignment, constraints are verified. If any constraint fails, Backtracking removes the assignment and tries another option.

Solution Development

Backtracking Execution Table

Step	Assignment	Constraint Check	Result
1	D1 → Morning	D1 ≠ Night	Accepted
2	D2 → Afternoon	Before D3	Accepted
3	D3 → Night	D3 ≠ Morning	Accepted
4	Unique shifts	No duplicates	Accepted
5	All doctors assigned	Complete	Goal State

Backtracking Tree



Final Schedule

Shift	Assigned Doctor
Morning	D1
Afternoon	D2
Night	D3

Application of Concepts

The following AI concepts are applied:

- Constraint Satisfaction Problem (CSP)
- Backtracking Search
- State Space Search
- Constraint Checking
- Goal State Testing

Justification

Backtracking Search is the most appropriate algorithm because it systematically explores possible assignments while eliminating invalid combinations immediately. This minimizes unnecessary computations and guarantees a valid schedule whenever one exists.

Result

A valid doctor schedule is successfully generated without violating any constraints.

Conclusion

The hospital scheduling problem was successfully solved using the Backtracking Search algorithm. All given constraints were checked at every step, and a valid schedule was obtained without any conflicts. The algorithm efficiently eliminated invalid assignments and produced the final schedule: D1 – Morning, D2 – Afternoon, and D3 – Night. This demonstrates how Backtracking effectively solves Constraint Satisfaction Problems (CSPs) by ensuring all constraints are satisfied.

QUESTION 2

Robot Navigation using Breadth First Search (BFS)

Rubric 1: Problem Understanding (25%)

Problem Statement

A robot must move from Start (S) to Goal (G) inside a 5×5 grid while avoiding obstacles. Each movement has equal cost (1), and only four-direction movement is allowed.

Grid Environment

```
S  □  □  X  □  
□  X  □  X  □  
□  □  □  □  □  
X  X  □  X  □  
□  □  □  G  □
```

Legend

- S → Start
- G → Goal

- $X \rightarrow$ Obstacle
- $\square \rightarrow$ Free Cell

Rubric 2: Constraint Analysis (25%)

Constraint	Effect
Move Up, Down, Left, Right	Restricts robot movement
Obstacle	Blocks path
Cost = 1	Equal edge weights
Manhattan Distance	Estimates closeness to Goal

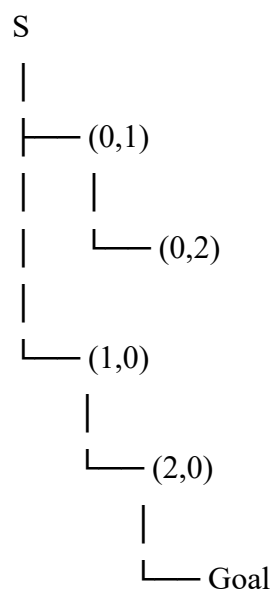
Because every move has equal cost, BFS guarantees the shortest path.

Solution Development

Node Expansion

Step	Expanded Node	Queue After Expansion
1	S	(0,1),(1,0)
2	(0,1)	(1,0),(0,2)
3	(1,0)	(0,2),(2,0)
4	(0,2)	(2,0),(1,2)
5	(2,0)	(1,2),(2,1)
6	Continue	Goal Fou

Search Flow:



Optimal Path

S

↓

↓

→

→

↓

↓

→

G

Cost Calculation

Number of Moves	Cost
7	$7 \times 1 = 7$

Application of Concepts

Applied AI Concepts

- Breadth First Search
- Graph Traversal
- Queue Data Structure
- State Space Search
- Shortest Path Search

Justification

Since all movement costs are identical, Breadth First Search guarantees the shortest path. BFS systematically explores nodes level by level, ensuring that the first path reaching the goal is optimal.

Result

The robot successfully reaches the goal with minimum path cost 7.

Conclusion

The robot navigation problem was successfully solved using the Breadth First Search (BFS) algorithm. By expanding nodes level by level, BFS found the shortest path from the Start node to the Goal while avoiding obstacles. Since each move had an equal cost, BFS guaranteed the

optimal solution with the minimum path cost. This experiment demonstrates the effectiveness of BFS in solving pathfinding and graph search problems.

Question 3

Autonomous Rescue Robot

Problem Understanding

Problem Statement

An autonomous rescue robot must safely navigate a disaster-affected building to rescue survivors. The robot operates in a partially observable environment, where blocked paths and risky zones are discovered dynamically during navigation.

Given Constraints

Constraint	Impact
Four-direction movement	Limited navigation
Obstacles	Cannot enter blocked rooms
Limited sensing	Partial environment knowledge
Normal movement cost = 1	Base travel cost
Risk zone cost = +2	Higher total cost
Online updates	Dynamic replanning

Rubric 2: Constraint Analysis

The robot must continuously evaluate newly observed information. Risky cells increase travel cost, while blocked cells force route changes. Therefore, decision-making depends on both current knowledge and newly sensed information.

Priority of Constraints:

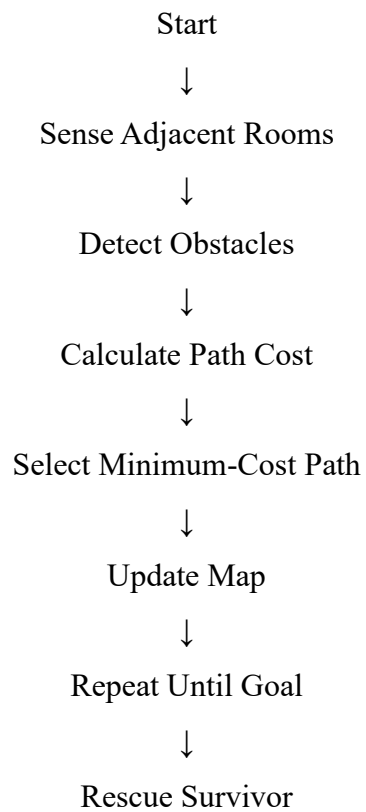
1. Avoid obstacles
2. Minimize total cost
3. Update environment knowledge
4. Reach survivor safely

Solution Development

Selected Strategy

Uniform Cost Search (UCS) + Online Search Agent

Execution Flow



Cost Formula

Normal Cell

Cost = Previous Cost + 1

Risky Cell

Cost = Previous Cost + 3

Application of Concepts

Intelligent Agent

The robot senses, plans, decides, and acts independently.

Rationality

The robot always selects the safest and least-cost path.

Environment Type

Property	Type
Observability	Partially Observable
Environment	Dynamic
Agent	Intelligent
Decision Process	Sequential

Justification

Uniform Cost Search is appropriate because it considers cumulative path cost rather than only path length. Combining UCS with an Online Search Agent enables the robot to adapt to newly discovered obstacles and risky zones, ensuring safe and cost-efficient navigation.

Result

The rescue robot successfully reaches the survivor by dynamically updating its knowledge and choosing the minimum-cost safe path.

Conclusion

The autonomous rescue robot problem was effectively addressed using Uniform Cost Search (UCS) combined with an Online Search Agent. The robot continuously updated its knowledge of the environment, avoided blocked paths, minimized travel costs, and safely reached the survivor. The solution demonstrated important AI concepts such as intelligent agents, rational decision-making, dynamic environment handling, and cost-based search, making it well suited for real-world disaster rescue applications.